

ModBand: Design of a Modular Headband for Multimodal Data Collection and Inference

Avinash Ajit Nargund*

anargund@ucsb.edu

University of California, Santa Barbara
Santa Barbara, California, USA

Arthur Caetano

caetano@ucsb.edu

University of California, Santa Barbara
Santa Barbara, California, USA

Alejandro Aponte*

aponte@ucsb.edu

University of California, Santa Barbara
Santa Barbara, California, USA

Misha Sra

sra@cs.ucsb.edu

University of California, Santa Barbara
Santa Barbara, California, USA



Figure 1: The figures show ModBand, our proposed low-cost, wearable 3D printed modular headband for face and physiological data collection and inference. Side, front and top views in sets of three from left to right: a) ModBand hardware, b) ModBand with sensors, and c) ModBand integrated with HoloLens 2.

ABSTRACT

Collecting multimodal user data during physical tasks such as cooking, maintenance, or physical rehab is crucial to enable the design of better AI models, interfaces, and applications. However, this is a challenging task with external cameras and sensors due to user movement, self-occlusions and diversity of data streams during task performance. In this work, we present ModBand, a wearable sensor headband with an accompanying software pipeline to collect and visualize data such as facial images, pupillometry, egocentric video, and heart rate during physical task performance. ModBand can be modified, extended, and used both as a standalone device or integrated with existing head-mounted AR devices for AI-based task guidance. Our modular design incorporates cost-effective fabrication methods, such as 3D printing, and enables convenient integration or exclusion of sensors to support custom data collection needs.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**;

KEYWORDS

Sensor headband, Wearable, Mobile, Modular design, Facial data collection, Visual and physiological data

*Both authors contributed equally to this research.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UIST '23 Adjunct, October 29–November 01, 2023, San Francisco, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0096-5/23/10.

<https://doi.org/10.1145/3586182.3616682>

ACM Reference Format:

Avinash Ajit Nargund, Alejandro Aponte, Arthur Caetano, and Misha Sra. 2023. ModBand: Design of a Modular Headband for Multimodal Data Collection and Inference. In *The 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23 Adjunct)*, October 29–November 01, 2023, San Francisco, CA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3586182.3616682>

1 INTRODUCTION

Wearable devices can collect multimodal data to describe a user's experience and behavior in a holistic way, which enables applications in various domains such as learning, health, and entertainment. There are several options to collect multimodal user data using consumer devices such as smartphones, smartwatches, and custom-built head-mounted or body-mounted devices. Research devices such as MeCap (camera-based egocentric body pose data) [1] and MonoEye (camera-based motion capture data) [8] allow egocentric tracking of a user's body, useful for extended reality (XR) applications. PhysioHMD (EEG, EMG, EDA, EOG, and PPG data for virtual reality (VR) headsets) [3] and its commercialized version Galea [2] are both VR headset add-ons that enable physiological data collection using an array of non-visual sensors. OpenSense (audio-video, speech, and facial expression data) [12] and EZ-MMLA (website for collecting audio-video data) [6] use visual sensors to collect user data though require the user to be seated at a desk. However, none of these devices support data collection in mobile conditions such as during cooking or performing maintenance or repair. These tasks present new challenges including user movement and self-occlusions (e.g., bending over so the face is no longer visible to an external camera). Since most prior devices were custom-built to support a specific research project, they are not available for others to build on, requiring researchers to often start from scratch if they need a data collection solution [4, 5, 10]. While Galea is available

for purchase starting at \$25k, it can be prohibitively expensive and it is not designed as a customizable device.

To overcome these limitations, we introduce ModBand, a low-cost modular headband for mobile multimodal user data collection through easily re-configurable 3D printed modules coupled with plug-and-play software to store, visualize, and synchronize collected data. ModBand can be used as a standalone device to collect data and it can also be integrated with existing head-mounted displays (HMDs) such as the Microsoft HoloLens¹ to give researchers, developers, and creators a powerful new tool for understanding and augmenting user-centered experiences. We have also developed a JavaScript-based graphical user application, *ModBand App* for realtime visualization of the data captured by all the sensors.

2 MODBAND DESIGN

ModBand design parameters were derived from an analysis of current devices for facial data collection: modularity, mobility, augmentation, and data accessibility. ModBand achieves modularity through interchangeable sections and expansion modules that can be attached to the headband. ModBand achieves mobility through its wearable form factor which allows all sensors to be connected to a single-board computer or both PC and mobile micro-computer. ModBand's design can augment a HoloLens 2 by transferring the external modules with the mounted sensors in a similar form factor. This enables consistency in the data collection across devices maintaining a similar sensor configuration. Lastly, the raw sensor data is handled in ROS² nodes and published in ROS messages. These messages can be used directly in applications or saved to disk for data analysis. Users of ModBand can arrange sensors and headband modules into a configuration that can support data collection for various tasks. All the data is time synchronized to enable easy correlations between attention, emotion, and activity recognition.

2.0.1 Fisheye Cameras (100 Hz). Although the images produced by this type of camera are distorted in a spherical format, they can capture information about a larger portion of the environment than conventional rectilinear cameras. ModBand can connect to fisheye cameras through USB and capture images at rates from 75 Hz to 100 Hz. For additional flexibility, ModBand can support outward- and inward-facing sensors with articulated arms attached to the lateral headband, allowing researchers to change the position and orientation of the cameras easily. This capability allows researchers to place fisheye cameras facing either inwards, to collect facial expressions, or outwards, to collect body poses. The video data stream can be visualized using the *ModBand App* and optionally saved to disk.

2.0.2 Heartrate Sensor (4Hz). In ModBand, we use an ambient light-based heartrate sensor from *PulseSensor* [9] for measuring the heartbeats per minute. The sensor is essentially a current-to-voltage converter. The output analog voltage of the sensor is converted to digital using the *ADS1015* ADC. The output of the ADC is sent to the SBC using I2C and can be visualized and stored.

2.0.3 IMU (50Hz). The IMU is connected to the SBC using I2C. Before integrating the IMU into the rest of the hardware setup it is calibrated using the procedure described in [7].

2.0.4 Pupil Core (125Hz). ModBand integrates not only raw sensors in a comprehensive platform but also third-party data collection platforms that provide minimum data sharing capabilities such as Pupil Core³ to collect, store, and time synchronize ocular data with the other data being collected.

3 DATA COLLECTION SOFTWARE

Data collection is done by synchronizing timestamps across different sensors and storing the incoming data on a disk. Since our modular device allows for using sensors in many different configurations, here we present one example configuration and the associated data pipeline. An overview of the data pipeline is shown in Figure 2. The pipeline consists of a Linux server and a single-board computer (SBC), specifically the Jetson Xavier NX. The Linux server acts as the central node in this network of sensors. The ROS master runs on the server, aggregates the data from different sensors, and stores them on disk. This system architecture provides a convenient method for conducting user studies as it makes ModBand wearable and portable.

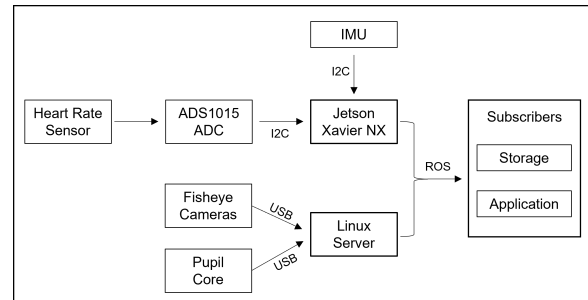


Figure 2: The ModBand data pipeline. The ROS messages can be stored or used by the ModBand app for data visualization.

4 LIMITATIONS AND FUTURE WORK

ModBand offers a novel approach for generating data and support to infer a user's emotional and cognitive states during physical task scenarios such as cooking, repair or rehab therapy. Our design presents several advantages over conventional data collection methods using consumer headsets or one-off research devices, such as replication, modularity, and extensibility. Our headband design makes data collection (standalone or integrated with a headset) more accessible, low-cost and mobile, without compromising the quality of data generated. An immediate next step to add flexibility is to save the data into ROS *bag* files which will allow us to use existing open-source tools, such as [11] to create datasets easily but replay the sensor data to debug applications. In the future, we plan on evaluating the data collection pipeline, building a dataset and testing the user interface.

¹<https://www.microsoft.com/en-us/hololens>

²<https://www.ros.org/>

³<https://pupil-labs.com/products/core/>

ACKNOWLEDGMENTS

The first version of this project was supported by DARPA HR0011-2220001 Subaward No. 505232-78050. We thank Sherry Chen for the original Python GUI, Mia Zhou for the Web-based GUI, and Kriteen Shrestha for helping with pilot data collection.

REFERENCES

- [1] Karan Ahuja, Chris Harrison, Mayank Goel, and Robert Xiao. 2019. MeCap: Whole-Body Digitization for Low-Cost VR/AR Headsets. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology* (New Orleans, LA, USA) (UIST '19). Association for Computing Machinery, New York, NY, USA, 453–462. <https://doi.org/10.1145/3332165.3347889>
- [2] Guillermo Bernal, Nelson Hidalgo, Conor Russomanno, and Pattie Maes. 2022. Galea: A physiological sensing system for behavioral research in Virtual Environments. In *2022 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. IEEE, 66–76.
- [3] Guillermo Bernal, Tao Yang, Abhinandan Jain, and Pattie Maes. 2018. PhysioHMD: a conformable, modular toolkit for collecting physiological data from head-mounted displays. In *Proceedings of the 2018 ACM International Symposium on Wearable Computers*. 160–167.
- [4] Yifeng Cao, Ashutosh Dhekne, and Mostafa Ammar. 2023. ViSig: Automatic Interpretation of Visual Body Signals Using On-Body Sensors. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7, 1 (2023), 1–27.
- [5] Alexander Curtiss, Blaine Rothrock, Abu Bakar, Nivedita Arora, Jason Huang, Zachary Enghardt, Aaron-Patrick Empedrado, Chixiang Wang, Saad Ahmed, Yang Zhang, et al. 2021. FaceBit: Smart face masks platform. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 5, 4 (2021), 1–44.
- [6] Javaria Hassan, Jovin Leong, and Bertrand Schneider. 2021. Multimodal Data Collection Made Easy: The EZ-MMLA Toolkit: A data collection website that provides educators and researchers with easy access to multimodal data streams. In *LAK21: 11th international learning analytics and knowledge conference*. 579–585. <https://doi.org/10.1145/3448139.3448201>
- [7] J. Hrisko. 2021. Gyroscope and Accelerometer Calibration with Raspberry Pi. *Maker Portal*. <https://makersportal.com/blog/calibration-of-a-magnetometer-with-raspberry-pi>
- [8] Dong-Hyun Hwang, Kohei Aso, Ye Yuan, Kris Kitani, and Hideki Koike. 2020. Monoeye: Multimodal human motion capture system using a single ultra-wide fisheye camera. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology*. 98–111.
- [9] WORLD FAMOUS ELECTRONICS LLC. 2023. *Pulse Sensor*. https://cdn.shopify.com/s/files/1/0100/6632/files/Pulse_Sensor_Data_Sheet.pdf?14358792549038671331
- [10] Johannes Meyer, Adrian Frank, Thomas Schlebusch, and Enkeljeda Kasneci. 2021. A cnn-based human activity recognition system combining a laser feedback interferometry eye movement sensor and an imu for context-aware smart glasses. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 5, 4 (2021), 1–24.
- [11] Southwest Research Institute Robotics. 2015. *Bag Database*. <https://github.com/swri-robotics/bag-database>
- [12] Kalin Stefanov, Baiyu Huang, Zongjian Li, and Mohammad Soleymani. 2020. OpenSense: A platform for multimodal data acquisition and behavior perception. In *Proceedings of the 2020 International Conference on Multimodal Interaction*. 660–664. <https://doi.org/10.1145/3382507.3418832>